SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)*

QUESTIONS: 05 **Total Marks: 100**

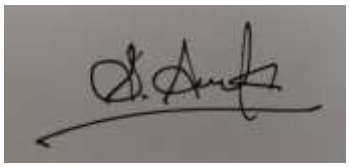
Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **Aravind kumar.S (192524210)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Question 1

1. A college wants to analyze student performance using attendance records, internal marks, assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.

Concept Map 1

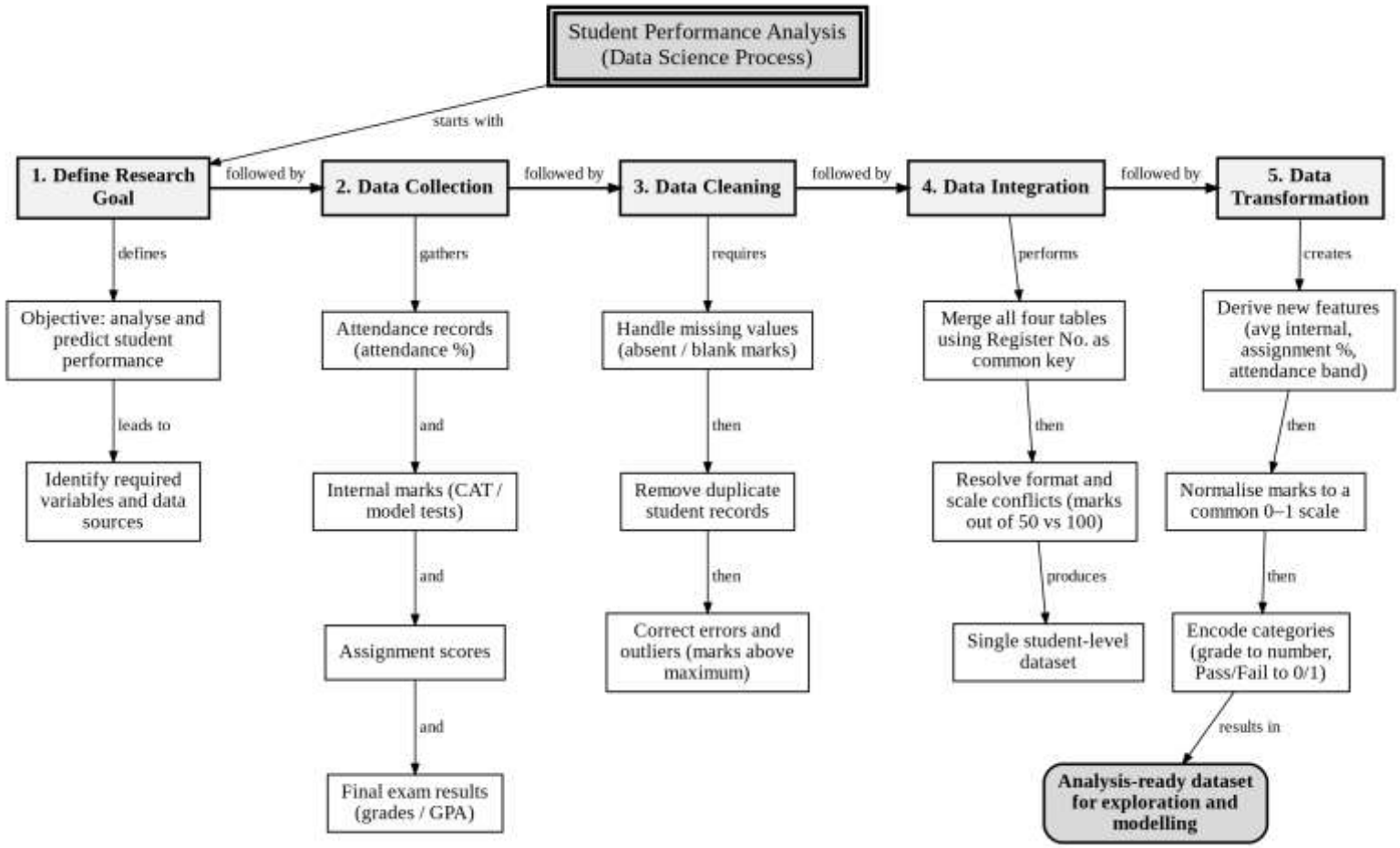


Figure 1. Concept map for Question 1

Explanation

The map follows the standard data science process: define the goal, collect data, clean it, integrate it, and transform it into a form suitable for analysis (Cielen, Meysman & Ali, 2016; Chapman et al., 2000). The college data comes from four sources, so the main work before analysis is to make these sources consistent and combine them at student level.

Concept	Role and linkage in the map
Define Research Goal	States why the data is needed (analyse and predict performance) and decides which variables to collect.
Data Collection	Gathers attendance, internal marks, assignment scores and final exam results from college records.
Data Cleaning	Fixes missing marks, duplicate records and impossible values so that later steps are reliable.
Data Integration	Joins the four tables using Register No. as the common key to get one student-level dataset.
Data Transformation	Creates derived features, scales marks and encodes categories to make the data model-ready.

Key linkages

- Each stage produces the output that becomes the input of the next stage, so the process is sequential but can be repeated when problems are found.
- Register No. links all four data sources and is the key relationship that makes integration possible.
- The data is mostly structured and numeric, so transformation focuses on scaling and derived variables.

Literature support

Cielen et al. (2016) describe the data science process as setting the goal, retrieving, preparing, exploring, modelling and presenting data. Romero and Ventura (2010) show that educational data mining depends on well-prepared student data before performance prediction is attempted. Together these support the order of stages used in this map.

Question 2

2. A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.

Concept Map 2

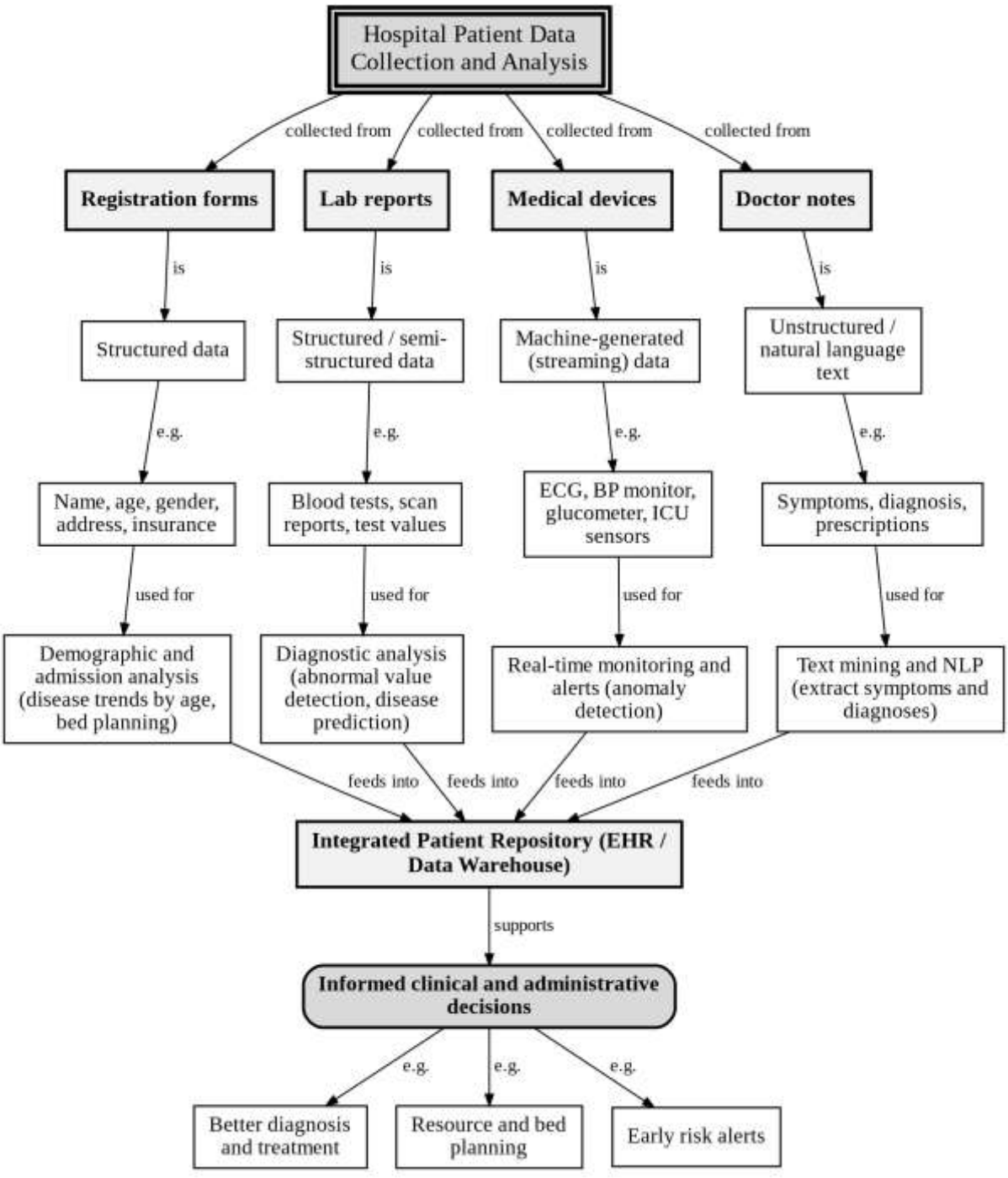


Figure 2. Concept map for Question 2

Explanation

The hospital data covers several facets of data: structured (registration forms, lab values), machine-generated (medical devices) and unstructured natural language (doctor notes) (Cielen et al., 2016). Each

source needs a different analysis approach, and the results are combined in one repository to support clinical and administrative decisions.

Concept	Role and linkage in the map
Registration forms	Structured demographic data used for admission analysis, disease trends by age and bed planning.
Lab reports	Structured or semi-structured test values used for diagnostic analysis and disease prediction.
Medical devices	Streaming sensor data (ECG, BP, glucose) used for real-time monitoring and anomaly alerts.
Doctor notes	Unstructured text that needs text mining or NLP to extract symptoms and diagnoses.
Integrated repository	EHR or data warehouse where all sources are combined for a complete patient view.

Key linkages

- The type of data decides the analysis method: structured data uses statistics and SQL, streaming data uses real-time analytics, and text uses NLP.
- All four branches meet in the integrated repository, which links every patient's records together.
- The final decisions (better treatment, resource planning, early alerts) depend on combining all sources, not on a single one.

Literature support

Raghupathi and Raghupathi (2014) explain that healthcare big data comes from many sources, in structured and unstructured form, and that its value appears when these sources are analysed together. This matches the structure of the map, where the four sources feed one repository.

Question 3

3. An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.

Concept Map 3

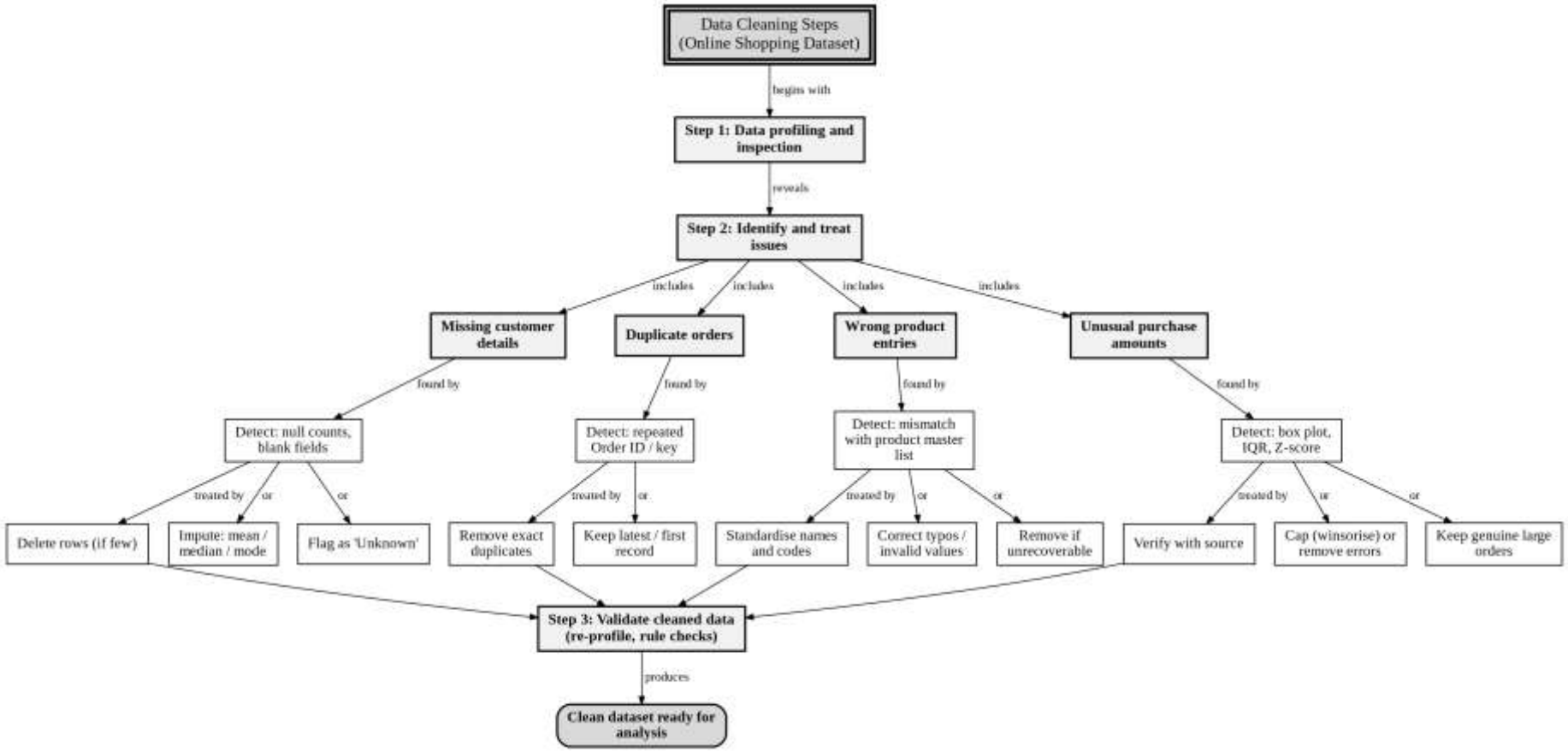


Figure 3. Concept map for Question 3

Explanation

Data cleaning starts with profiling to find the problems, then each problem is detected and treated with a suitable method, and finally the cleaned data is validated (Rahm & Do, 2000; Han, Kamber & Pei, 2012). The four problems in the shopping dataset are missing values, duplicates, invalid entries and outliers.

Concept	Role and linkage in the map
Missing customer details	Detected using null counts; treated by deletion, imputation (mean, median, mode) or flagging.
Duplicate orders	Detected using repeated Order ID or key; treated by removing exact duplicates and keeping one record.
Wrong product entries	Detected by comparing with the product master list; treated by standardising or correcting values.
Unusual purchase amounts	Detected using box plot, IQR or Z-score; verified with the source, then capped, removed or kept.
Validation	Re-checks the dataset after cleaning so that the data is ready for analysis.

Key linkages

- Profiling reveals the problems, so it must come before treatment.
- Each problem has its own detection method and treatment options, and the choice depends on how much data is affected.
- An outlier is not always an error, so large genuine orders are kept after verification.

Literature support

Rahm and Do (2000) classify data quality problems such as missing values, duplicates and invalid entries and describe detection and correction steps for each. Han et al. (2012) list the same methods (imputation, outlier detection, smoothing) under data cleaning, which are used in this map.

Question 4

4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.

Concept Map 4

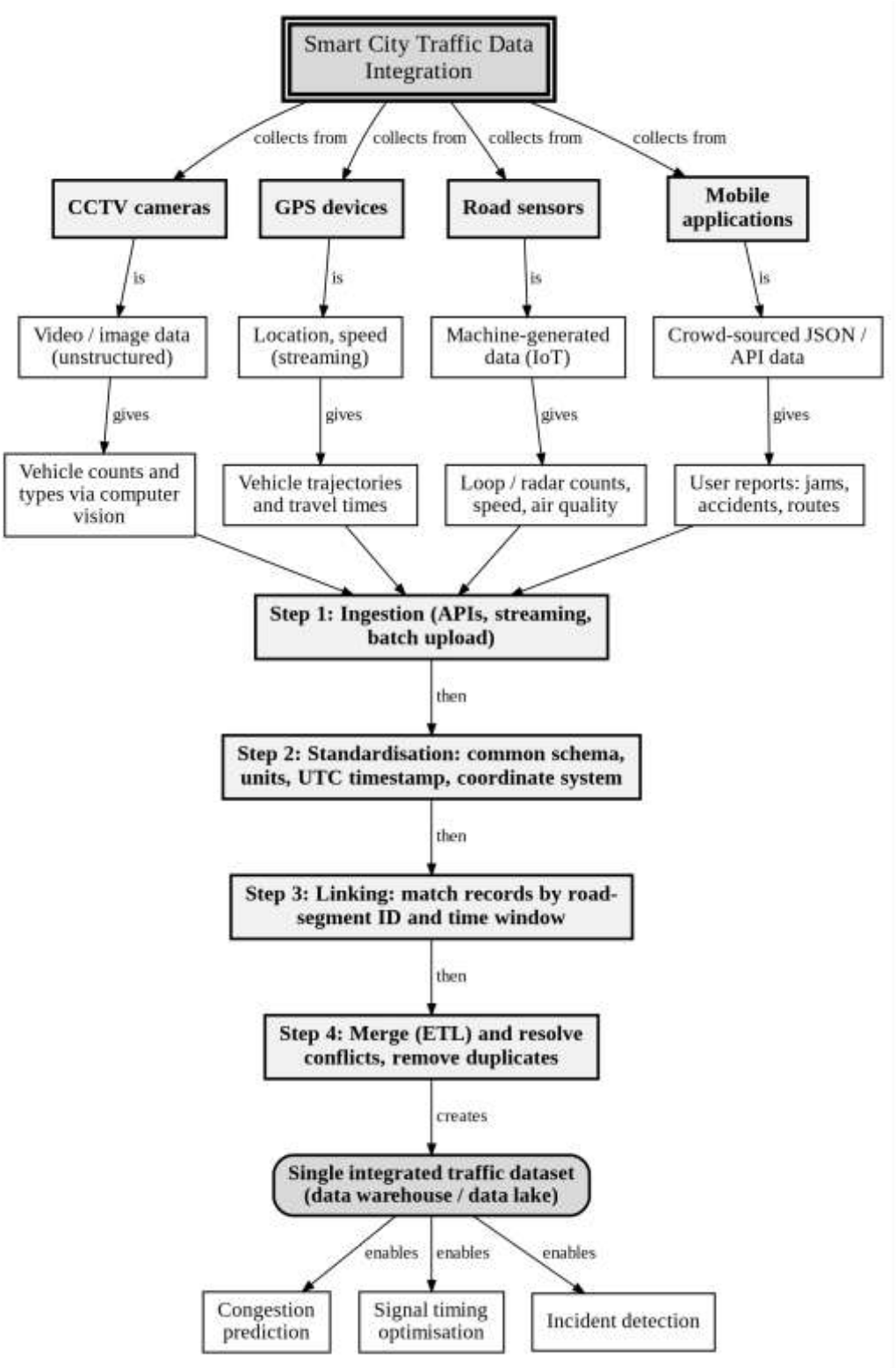


Figure 4. Concept map for Question 4

Explanation

The four sources differ in format, speed and structure: video from CCTV, streaming location data from GPS, machine-generated sensor readings and crowd-sourced app data. Integration therefore needs ingestion, standardisation to a common schema, linking of records, and merging with conflict resolution (Doan, Halevy & Ives, 2012).

Concept	Role and linkage in the map
CCTV cameras	Unstructured video; vehicle counts and types are extracted using computer vision.
GPS devices	Streaming location and speed data that gives trajectories and travel times.
Road sensors	Machine-generated IoT readings such as traffic counts, speed and air quality.
Mobile applications	Crowd-sourced reports in JSON or API form on jams, accidents and routes.
Integrated dataset	Single dataset in a data warehouse or data lake used for congestion prediction and signal planning.

Key linkages

- Standardisation of schema, units, time zone and coordinates makes records from different sources comparable.
- Road-segment ID and a time window act as the linking keys between sources.
- Conflicts and duplicate observations (for example a vehicle seen by both CCTV and a sensor) are resolved during merging.

Literature support

Doan et al. (2012) describe data integration as schema matching, entity matching and merging with conflict resolution. Zanella et al. (2014) describe smart city systems as many heterogeneous IoT data sources that must be brought together for urban services. Both support the stepwise flow shown in the map.

Question 5

5. A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

Concept Map 5

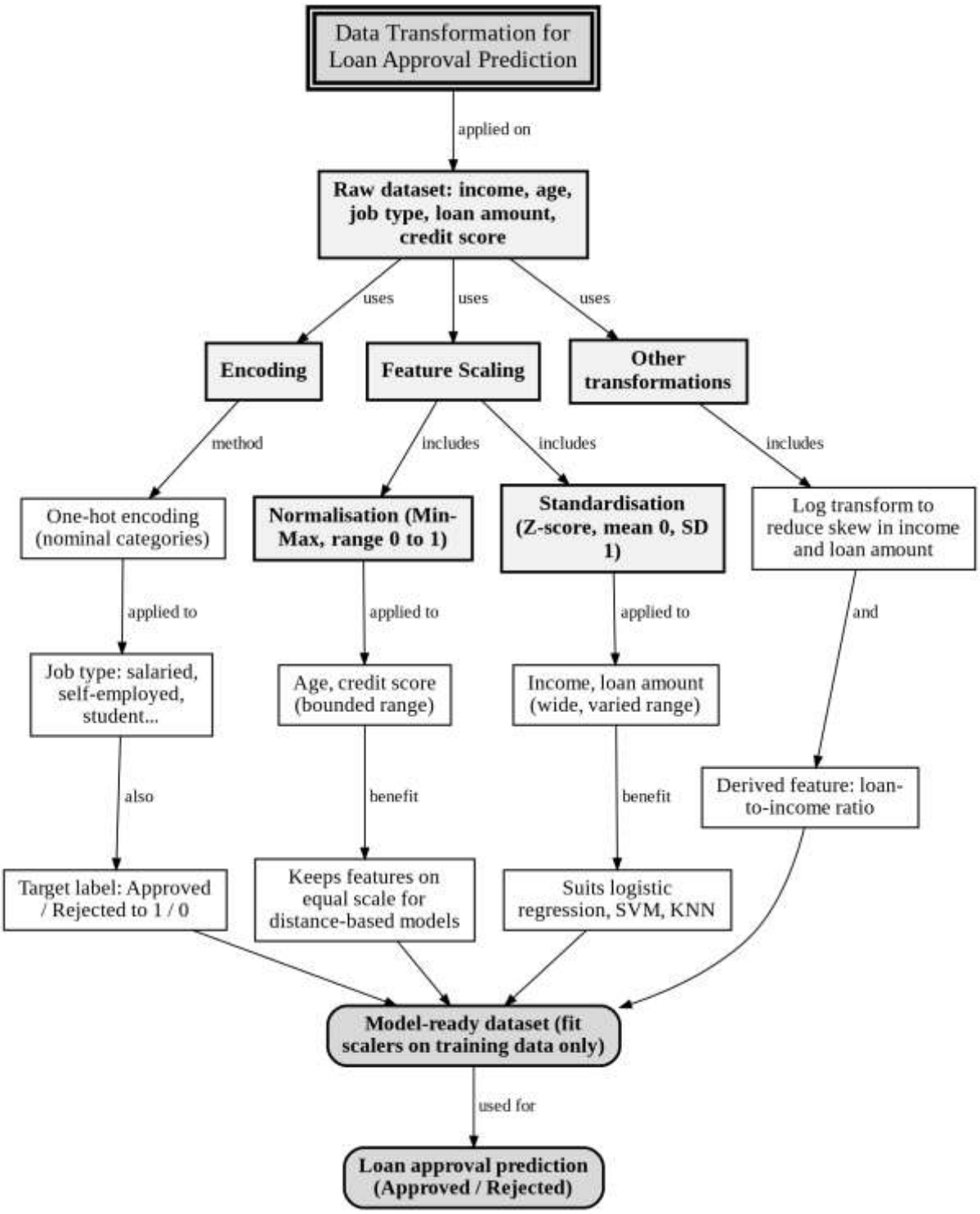


Figure 5. Concept map for Question 5

Explanation

The dataset has one categorical attribute (job type) and four numeric attributes with very different ranges (García, Luengo & Herrera, 2015). Encoding is used for the categorical attribute, normalisation and standardisation (both types of feature scaling) are used for numeric attributes, and extra transformations reduce skew and add useful features.

Concept	Role and linkage in the map
Encoding	One-hot encoding converts job type into binary columns; the Approved / Rejected label is coded as 1 / 0.
Normalisation (Min-Max)	Rescales age and credit score to the range 0 to 1.
Standardisation (Z-score)	Rescales income and loan amount to mean 0 and standard deviation 1; suits logistic regression, SVM and KNN.
Log transform	Reduces right skew in income and loan amount.
Derived feature	Loan-to-income ratio gives the model a direct measure of repayment burden.

Key linkages

- Feature scaling is the general technique, and normalisation and standardisation are its two common forms.
- Scalers are fitted on training data only and then applied to test data, to avoid data leakage.
- All transformed attributes together form the model-ready dataset used for loan approval prediction.

Literature support

García et al. (2015) present encoding of categorical variables and scaling of numeric variables as standard data preprocessing steps. Lessmann et al. (2015) benchmark credit scoring classifiers on preprocessed data, which shows that preparation of attributes such as income and credit score is part of loan prediction work.

References

Chapman, P., Clinton, J., Kerber, R., et al. (2000). CRISP-DM 1.0: Step-by-step data mining guide. SPSS Inc.

Cielen, D., Meysman, A., & Ali, M. (2016). Introducing Data Science. Manning Publications.

Doan, A., Halevy, A., & Ives, Z. (2012). Principles of Data Integration. Morgan Kaufmann.

García, S., Luengo, J., & Herrera, F. (2015). Data Preprocessing in Data Mining. Springer.

Han, J., Kamber, M., & Pei, J. (2012). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann.

Lessmann, S., Baesens, B., Seow, H.-V., & Thomas, L. C. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring. European Journal of Operational Research, 247(1), 124–136.

Raghupathi, W., & Raghupathi, V. (2014). Big data analytics in healthcare: Promise and potential. Health Information Science and Systems, 2, 3.

Rahm, E., & Do, H. H. (2000). Data cleaning: Problems and current approaches. IEEE Data Engineering Bulletin, 23(4), 3–13.

Romero, C., & Ventura, S. (2010). Educational data mining: A review of the state of the art. IEEE Transactions on Systems, Man, and Cybernetics, Part C, 40(6), 601–618.

Zanella, A., Bui, N., Castellani, A., Vangelista, L., & Zorzi, M. (2014). Internet of Things for smart cities. IEEE Internet of Things Journal, 1(1), 22–32.